

Project Report

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Task 1.

The estimation values of all, urban and rural populations for 2025 was calculated by utilizing the data from 2021 to 2024. Lagrange and Cubic Spline interpolations are taken as models to predict these values and the results are shown in the following table:

Table 1. Population Estimates for 2025 using Lagrange and Cubic Spline models

2025	Value from Dataset	Lagrange Interpolation	Cubic Spline Interpolation
All Population	20,286,084	20,667,610	20,300,877
Urban Population	12,774,743	13,358,968	12,692,488
Rural Population	7,511,341	7,308,642	7,608,389

The figure 1 below shows the graph of Lagrange polynomials:

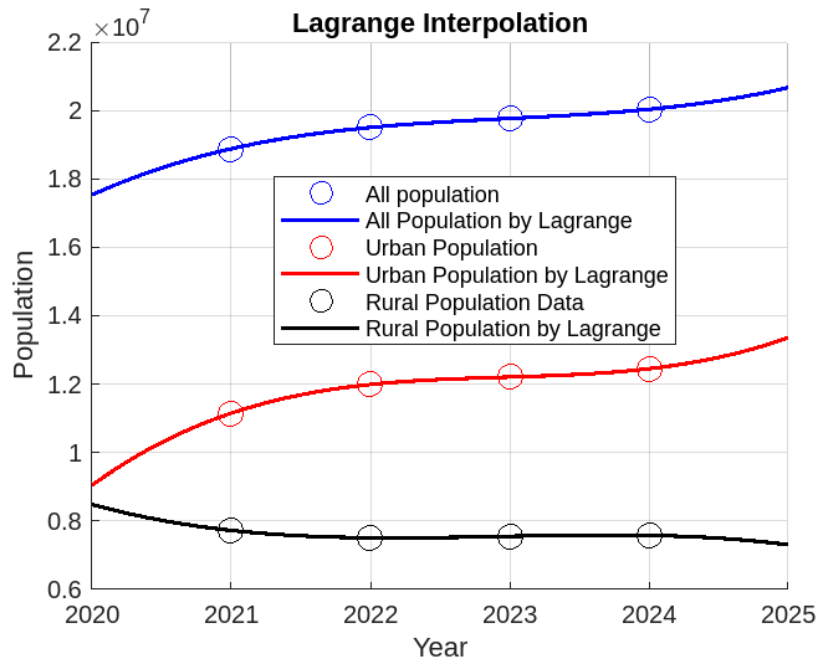


Figure 1. Lagrange Polynomial

The figure 2 shows Cubic spline polynomials. Both figures were obtained from Matlab working space.

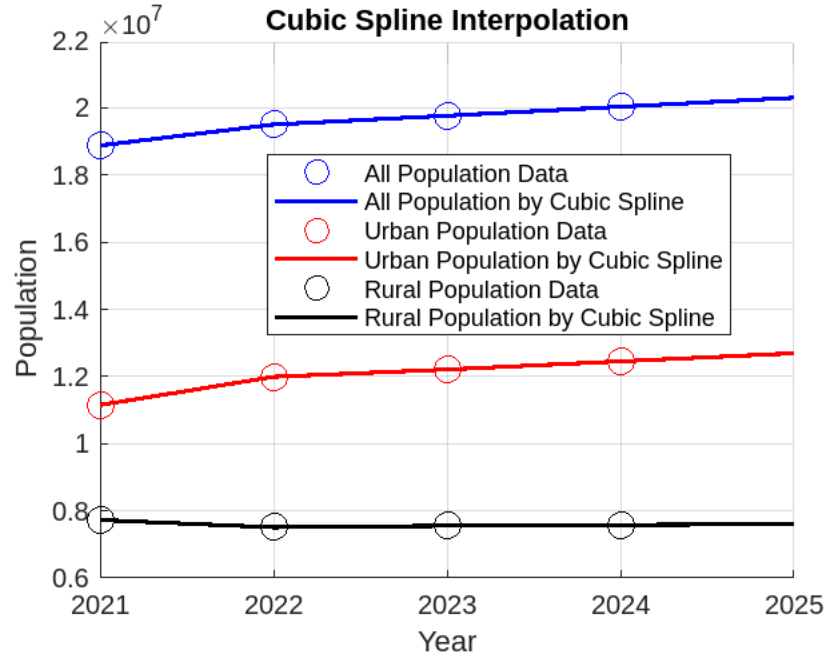


Figure 1. Cubic Spline Polynomial

Task 2.

The **absolute error** in v approximating u , can be calculated as follows:

$$|u - v| \quad (1)$$

The **relative error** (assuming $u \neq 0$) is

$$\frac{|u-v|}{|u|} \quad (2)$$

Using formulas (1) and (2), absolute and relative errors for each model predictions were calculated and the results as following:

Table 2. Absolute and Relative Errors for Lagrange Estimations

	Absolute Error	Relative Error
All Population	381,526	0.01881
Urban Population	584,225	0.04573
Rural Population	202,699	0.02699

Table 3. Absolute and Relative Errors for Cubic Spline Estimations

	Absolute Error	Relative Error
All Population	14,793	0.00073
Urban Population	82,255	0.00644
Rural Population	97,048	0.01292

Task 3.

Cubic spline interpolation is the most suitable method for this task compared to Lagrange interpolation. By comparing the results obtained in Tables 2 and 3, one can see that absolute and relative errors of cubic spline are lower than the Lagrange interpolation method. For example, according to Tables 2 and 3, the relative error of the Lagrange is 0.01881, whereas this value for cubic spline is only 0.00073. Similarly, for urban population, Lagrange's relative error is 0.04598, while cubic spline's error is only 0.00644 and for rural population, Lagrange has 0.02699 relative error and cubic spline has 0.01292. In all cases, the absolute and relative errors of the cubic spline are significantly smaller than the Lagrange interpolation and these show that the cubic spline better approximates the population values for 2025. Despite the fact that in both cases the number of points used was the same, there were some differences in approximation due to the technique that each model used. The Lagrange interpolation uses only one single high-degree polynomial, which might lead to the huge differences from the true value and make it unsuitable for predicting future results. On the other hand, the cubic spline method divides the given data points into ranges and uses 3-rd order polynomials for each range to construct a piecewise polynomial function further to approximate the data point. Also, it guarantees that the first and the second order derivatives are continuous for all ranges. As a result, the obtained function is smooth enough to predict values. So, it can be concluded that the cubic spline method is the most suitable method for this task, compared to Lagrange interpolation, as it produced the optimal population estimate value for 2025 dataset.

Task 4. When Kazakhstan's Population Will Reach 20.4 Million?

To determine when the population of Kazakhstan will reach 20.4 million, we will use the cubic spline interpolation method since it provides a more accurate and smoother outcome than Lagrange interpolation.

We construct a natural cubic spline interpolation model using all data points for the years 2021-2025. It consists of four piecewise polynomials, each one valid in a year interval:

- [2021, 2022]

$$f(x) = 18879552 + 720531(x - 2021) - 0(x - 2021)^2 + 96924(x - 2021)^3$$

- [2022, 2023]

$$f(x) = 19503159 + 429760(x - 2022) - 290771(x - 2022)^2 + 124659(x - 2022)^3$$

- [2023, 2024]

$$f(x) = 19766807 + 222196(x - 2023) - 83207(x - 2023)^2 + 38368(x - 2023)^3$$

- [2024, 2025]

$$f(x) = 20033842 + 273506(x - 2024) - 31897(x - 2024)^2 + 10632(x - 2024)^3$$

To calculate when the population reaches 20.4 million, we use $spline(x) = 20400000$ and solve for x using MATLAB: $x = 2025.7868$

Converting decimals to months: $0.7868 \times 12 \approx 9.44 \rightarrow$ mid-September
The population of Kazakhstan will reach 20.4 million in **mid-September of 2025**.

Task 5. What Are Alternative Population Prediction Methods?

In our previous tasks, we used Lagrange and Cubic Spline Interpolation to estimate the population changes in Kazakhstan. We calculated their accuracy and smoothness and concluded that the Cubic spline method is more accurate. However, there are a lot of other methods to predict population growth. Three alternative methods derived from journal articles used in different countries will be introduced and discussed in this task.

The first method is called ARIMA, which stands for AutoRegressive Integrated Moving Average. It is a time series model that uses past population data to predict future values. A study from Pakistan used ARIMA to forecast the population until 2027 using data from 1951. They found it to be accurate and reliable, especially for short-term changes. The model looks at trends and patterns in the past and uses them to make predictions. However, it requires the availability of past data and is sensitive to unexpected changes in demography shifts and government policy. [\[1\]](#)

The second method is a logistic-based grey forecasting model. It combines the logistic growth model and the grey system theory. The logistic part helps when population growth slows down over time, and the grey part helps when the data is uncertain or incomplete. It is particularly suitable for long-term predictions in environments where perfect data is not available. Recent research (Population Growth Prediction, 2023) confirmed that this method improves accuracy over standard logistic regression. It is also computationally simpler than machine learning models and more interpretable. However, it assumes that the population will reach a limit, which may not always be true. [\[2\]](#)

The third method uses machine learning. One study tested models like linear regression and gradient boosting on population data from 262 countries spanning 1960-2017. The study concluded that this model produced the most accurate population forecasts. They are suitable for large-scale forecasting across different countries. However, these models typically require extensive data, computing power, and tuning. Machine learning can deal with very complex data and changing conditions. It can be very accurate, but it needs a lot of data and expert setup. [\[3\]](#)

After comparing the three methods, our group believes that ARIMA is the most suitable model for forecasting Kazakhstan's population. It is simpler than machine learning, more realistic than logistic growth, and works well if historical data is available. This method can also be updated regularly, and that makes it a reliable method for both short and long term purposes.

Task 6.

To project the population in 2027, we used Model 2 (cubic spline interpolation). The estimated number of rural population in 2027 was found to be approximately 7 368 723. This result was derived from the data about a rural population from 2021 to 2025. Spline interpolation is an appropriate method in this problem for short-term extrapolation.

According to data, over the years 2021 to 2025, there was a decline in the rural population. This estimation reflects the real situation in Kazakhstan where people are prone to move to urban areas for

better quality of life and opportunities. The major reasons for urban migration are lack of job opportunities, poor infrastructure, and non-developing healthcare and education sector.

For investors, we would suggest the development of rural infrastructure and services to reduce the outflow of residents and at the same time attract urban residents. It is crucial to take advantage of the strengths of rural areas and improve them. By investing money in agricultural production, we can increase production and quality of products, as well as offer exclusive and extremely favorable price offers for residents to attract them financially. Moreover, the potential direction is investment in digital technologies and internet accessibility will also open possibilities for e-commerce and digital entrepreneurship in rural settings. By improving infrastructure, building hospitals, roads, and schools, and attracting valuable human resources from the cities through lucrative contracts we can attract more residents.

Task 7.

To project the number of population in 2027 we also used Model 2 (cubic spline interpolation). The estimated number of urban population in 2027 was found to be 13 421 845. The results were also obtained from urban population data from 2021 to 2025. Spline interpolation is an appropriate method in this problem for short-term extrapolation.

Evaluating the data for the period 2021-2025, we can conclude that the population has increased significantly in contrast to the rural population. Such changes are in direct contrast to trends in urban areas, as many residents of Kazakhstan are fleeing for better opportunities in the city, which has a much more developed infrastructure, quality education, and access to modern medical services. In addition, cities offer a broader labor market, career opportunities, and diversity in lifestyle, which is especially attractive to the younger generation. These factors contribute to the steady growth of the urban population, while rural areas continue to lose population.

For investors, we would suggest simultaneously investing and preventing the problem of overpopulation, namely by building energy-efficient houses with sustainable materials that can accommodate more residents without harming the environment. It is also possible to target another strategy to improve the transportation system of cities by opening up new opportunities, and introducing smart traffic lights and traffic control technologies. In addition, investments in education and innovation hubs can be an important area of focus, which will help develop human capital and create new jobs in cities.

References

1. Zakria, M., & Muhammad, F. (2009). *Forecasting the population of Pakistan using ARIMA models*. Pakistan Journal of Agricultural Sciences. <https://www.researchgate.net/publication/228468254>
2. Population Growth Prediction Using Logistic-Based Grey Forecasting Model. (2023). <https://www.researchgate.net/publication/384430896>
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